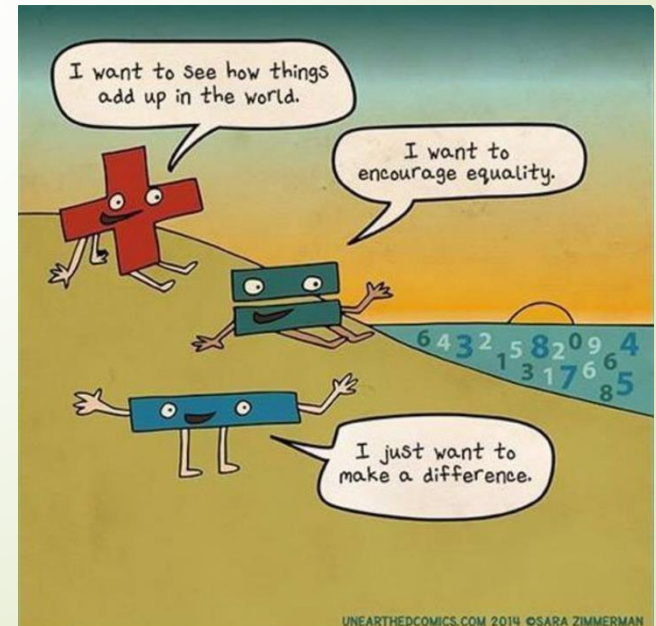
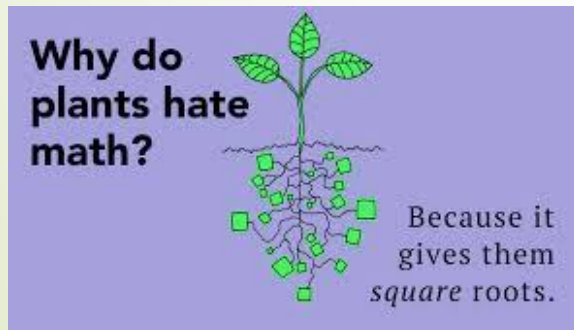


11Ext: 2 - Reviewing Surds

Lesson Objectives:

1. Simplify surds
2. Multiply and divide surds
3. Rationalise the denominator for surd expressions
4. Simplify Surd expressions involving n th roots, addition and subtraction



Recall: Surds are a convenient way to write numbers with lots of decimal places that have no pattern ie: they are ***irrational (I)***

Example: Instead of writing 2.236068....., we write $\sqrt{5}$

A surd is an exact value and is not rounded, therefore it is very accurate

A surd is a root that has no pattern and cannot be written as a whole number or a fixed decimal

Simplifying Surds

Examples:

$$\sqrt{45}$$

$$= \sqrt{3 \cdot 3 \cdot 5}$$

$$= 3\sqrt{5}$$

$$\sqrt[4]{32x^4y^5}$$

$$= \sqrt[4]{2 \cdot 2 \cdot 2 \cdot 2 \cdot x^4 \cdot y^4 \cdot y^1}$$

$$= 2xy \cdot \sqrt[4]{2y}$$

$$(2\sqrt{5})^3$$

$$= 2\sqrt{5} \times 2\sqrt{5} \times 2\sqrt{5}$$

$$= 2 \times 2 \times 2 \times \sqrt{5} \times \sqrt{5} \times \sqrt{5}$$

$$= 8 \times 5 \times \sqrt{5}$$

$$= 40\sqrt{5}$$

$$\sqrt[3]{24m^3}$$

$$\sqrt[3]{1000}$$

$$\sqrt[3]{-162}$$

$$\sqrt[4]{128x^7y^7}$$

Multiplication & Division

Use the following rules:

$$\sqrt{ab} = \sqrt{a} \times \sqrt{b} \quad \text{and} \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Examples:

$$\sqrt{18x} \cdot \sqrt{2x}$$

$$= \sqrt{36x^2}$$

$$= 6x$$

$$\sqrt[3]{16x^2y} \cdot \sqrt[3]{2xy^2z}$$

$$= \sqrt[3]{32x^3y^3z}$$

$$= \sqrt[3]{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot x^3 \cdot y^3 \cdot z}$$

$$= 2xy \cdot \sqrt[3]{4z}$$

Examples:

$$\frac{\sqrt[3]{16}}{\sqrt[3]{2}}$$

$$= \sqrt[3]{\frac{16}{2}}$$

$$= \sqrt[3]{8}$$

$$= 2$$

$$\frac{\sqrt{9xy^5}}{\sqrt{27x^5y^3}}$$

$$= \sqrt{\frac{9xy^5}{27x^5y^3}}$$

$$= \sqrt{\frac{xy^5}{3x^5y^3}}$$

$$= \sqrt{\frac{x^{-4}y^2}{3}}$$

$$= \sqrt{\frac{y^2}{3x^4}}$$

$$= \frac{\sqrt{y^2}}{\sqrt{3 \cdot x \cdot x \cdot x \cdot x}}$$

$$= \frac{y}{x^2\sqrt{3}}$$

Simplify

$$\sqrt[3]{5a^2} \cdot \sqrt[3]{2a}$$

$$\sqrt[3]{s^2t^4} \cdot \sqrt[3]{s^4t^6}$$

$$\frac{6\sqrt{45x^3}}{3\sqrt{5x}}$$

$$\frac{9\sqrt[5]{160x^8y^{11}}}{3\sqrt[5]{5xy^2}}$$

Simplifying surds involving nth roots

For $a, b \geq 0$: $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$; for $a \geq 0, b > 0$: $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$.

Practice 4

1 Write as a single power of 2:

a $8\sqrt{2}$

b $4\sqrt[3]{2}$

c $\frac{16}{\sqrt{2}}$

d $\frac{\sqrt[3]{2}}{4}$

1 a $2^{\frac{7}{2}}$

b $2^{\frac{7}{3}}$

c $2^{\frac{7}{2}}$

d $2^{-\frac{5}{3}}$

Objective C: Communicating

iii. move between different forms of mathematical representation

In Q4, write radical expressions as exponents to simplify them.

- 4** Write these expressions as powers of prime numbers. Use the rules of exponents to simplify. Give your answers as powers of prime numbers.

a $\sqrt{2} \times \sqrt[3]{2}$

b $\sqrt[3]{9} \times \sqrt[4]{9}$

c $\sqrt[5]{7} \times \sqrt[10]{7}$

d $\frac{\sqrt{6}}{\sqrt[3]{6}}$

4 a $2^{\frac{5}{6}}$

b $3^{\frac{7}{6}}$

c $7^{\frac{3}{10}}$

d $2^{\frac{1}{6}} 3^{\frac{1}{6}}$

When radicals have the same radicand but different indices, write them with fractional exponents to simplify them. For example:

$$\bullet \sqrt[n]{a} \times \sqrt[m]{a} = a^{\frac{1}{n}} \times a^{\frac{1}{m}} = a^{\frac{n+m}{nm}} \quad \bullet \frac{\sqrt[n]{a}}{\sqrt[m]{a}} = \frac{a^{\frac{1}{n}}}{a^{\frac{1}{m}}} = a^{\frac{1}{n} - \frac{1}{m}} = a^{\frac{n-m}{nm}}$$

The radicand is the number under the square root sign.

ATL Example 5

Simplify $\frac{\sqrt[4]{8}}{\sqrt{2}}$

$$\frac{\sqrt[4]{8}}{\sqrt{2}} = \frac{\sqrt[4]{2^3}}{\sqrt{2}}$$

Write the radicands as exponents with the same base.

$$= \frac{2^{\frac{3}{4}}}{2^{\frac{1}{2}}}$$

$$= 2^{\frac{3}{4} - \frac{1}{2}}$$

$$= 2^{\frac{1}{4}}$$

Give the answer as a radical, with prime number radicand.

$$= \sqrt[4]{2}$$

ATL Example 6

Write $10\sqrt{5} \times 3\sqrt[3]{75}$ as a simplified radical, by expressing the radicands as powers of prime numbers.

$$\begin{aligned}10\sqrt[3]{5} \times 3\sqrt[3]{75} &= 10 \times \sqrt[3]{5} \times 3 \times \sqrt[3]{25 \times 3} \\&= 10 \times 3 \times \sqrt[3]{5} \times \sqrt[3]{5^2} \times \sqrt[3]{3} \\&= 30 \times 5^{\frac{1}{3}} \times 5^{\frac{2}{3}} \times 3^{\frac{1}{3}} \\&= 30 \times 5^1 \times 3^{\frac{1}{3}} \\&= 150\sqrt[3]{3}\end{aligned}$$

Write the radicands as multiples of primes.

Write radicands using exponents.

Practice 5

1 Simplify these completely, leaving your answer as a simplified radical.

a $\sqrt{2} \times \sqrt[3]{2} \times \sqrt[6]{2}$ **b** $\sqrt[3]{3^2} \times \sqrt[4]{3} \times \sqrt[12]{3}$

f $\frac{\sqrt[3]{3^4}}{\sqrt{3}}$ **g** $3\sqrt{2} \times 5\sqrt[3]{2}$

2 Write as simplified radicals, with prime number radicands.

a $\frac{4}{\sqrt[3]{2}}$

b $\sqrt[4]{12} \times \sqrt[3]{3}$

f $\frac{\sqrt[4]{125}}{\sqrt[3]{5}}$

g $\frac{4\sqrt{6}}{\sqrt{3}}$

Problem solving

3 Solve these equations, where the unknown is always a power of 3.

a $\frac{\sqrt{27}}{x} = \sqrt{3}$

b $\frac{y}{3} = \frac{1}{\sqrt[3]{9}}$

Practice 5

1 Simplify these completely, leaving your answer as a simplified radical.

a $\sqrt{2} \times \sqrt[3]{2} \times \sqrt[6]{2}$

b $\sqrt[3]{3^2} \times \sqrt[4]{3} \times \sqrt[12]{3}$

f $\frac{\sqrt[3]{3^4}}{\sqrt{3}}$

g $3\sqrt{2} \times 5\sqrt[3]{2}$

a) $2^{\frac{1}{2}} \cdot 2^{\frac{1}{3}} \cdot 2^{\frac{1}{6}} = 2^{\frac{2}{6} + \frac{2}{6} + \frac{1}{6}} = 2^1 = \boxed{2}$

b) $\boxed{3}$

f) $\frac{\sqrt[3]{3^4}}{\sqrt{3}} = \frac{3^{\frac{4}{3}}}{3^{\frac{1}{2}}} = 3^{\frac{4}{3} - \frac{1}{2}} = 3^{\frac{5}{6}} = \boxed{\sqrt[6]{3^5}}$

g) $\boxed{15 \cdot \sqrt{2^5}}$

2 Write as simplified radicals, with prime number radicands.

a $\frac{4}{\sqrt[3]{2}}$

b $\sqrt[4]{12} \times \sqrt[3]{3}$

f $\frac{\sqrt[4]{125}}{\sqrt[3]{5}}$

g $\frac{4\sqrt{6}}{\sqrt{3}}$

a) $\frac{2^2}{2^{\frac{1}{3}}} = 2^{2 - \frac{1}{3}} = 2^{\frac{5}{3}} = \boxed{\sqrt[3]{2^5}}$

b) $\sqrt[4]{12} \cdot \sqrt[3]{3} = \sqrt[4]{2 \cdot 2 \cdot 3} \cdot \sqrt[3]{3} = 2^{\frac{1}{4}} \cdot 2^{\frac{1}{4}} \cdot 3^{\frac{1}{4}} \cdot 3^{\frac{1}{3}} = 2^{\frac{1}{2}} \cdot 3^{\frac{1}{4} + \frac{1}{3}} = \sqrt{2} \cdot 3^{\frac{7}{12}} = \boxed{\sqrt{2} \cdot \sqrt[12]{3^7}}$

Problem solving

3 Solve these equations, where the unknown is always a power of 3.

a $\frac{\sqrt{27}}{x} = \sqrt{3}$

b $\frac{y}{3} = \frac{1}{\sqrt[3]{9}}$

$y = \sqrt[3]{3}$

$\frac{\sqrt{27}}{x} = \sqrt{3}$

$\frac{x}{\sqrt{3}} = x$

$\frac{3^{\frac{1}{2}}}{3^{\frac{1}{2}}} = x$

$x = 3$

f) $\boxed{\sqrt[12]{5^5}}$

g) $\boxed{4\sqrt{2}}$

Rationalising the denominator

When an expression involves division by a surd, we can write the expression with an **integer denominator** (one that does **not** contain surds).

To do this we use the rule $\sqrt{a} \times \sqrt{a} = a$.

For example: $\frac{6}{\sqrt{3}}$ can be written as $\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ where we are really multiplying the original fraction by 1 (in the form $\frac{\sqrt{3}}{\sqrt{3}}$).

$\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ then simplifies to $\frac{6\sqrt{3}}{3}$ or $2\sqrt{3}$.

$$\begin{aligned} & \frac{10}{\sqrt{5}} \\ = & \frac{10}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} \\ = & \frac{10}{5} \sqrt{5} \\ = & 2\sqrt{5} \end{aligned}$$

$$\frac{10}{2\sqrt{2}}$$

Rationalising with a binomial denominator

Example 14

Self Tutor

Express $\frac{1}{3 + \sqrt{2}}$ with integer denominator.

$$\begin{aligned}\frac{1}{3 + \sqrt{2}} &= \left(\frac{1}{3 + \sqrt{2}} \right) \left(\frac{3 - \sqrt{2}}{3 - \sqrt{2}} \right) \\ &= \frac{3 - \sqrt{2}}{3^2 - (\sqrt{2})^2} \quad \{\text{using } (a + b)(a - b) = a^2 - b^2\} \\ &= \frac{3 - \sqrt{2}}{7}\end{aligned}$$

We are really multiplying by one, which does not change the value of the original expression.



Exercise 2:

EXERCISE 3D

1 Express with integer denominator:

a $\frac{1}{\sqrt{2}}$

b $\frac{2}{\sqrt{2}}$

c $\frac{4}{\sqrt{2}}$

f $\frac{1}{\sqrt{3}}$

g $\frac{3}{\sqrt{3}}$

h $\frac{4}{\sqrt{3}}$

k $\frac{1}{\sqrt{5}}$

l $\frac{3}{\sqrt{5}}$

m $\frac{\sqrt{3}}{\sqrt{5}}$

p $\frac{\sqrt{10}}{\sqrt{2}}$

q $\frac{1}{2\sqrt{3}}$

r $\frac{2\sqrt{2}}{\sqrt{3}}$

2 Express with integer denominator:

a $\frac{1}{3 - \sqrt{5}}$

b $\frac{1}{2 + \sqrt{3}}$

e $\frac{\sqrt{3}}{3 + \sqrt{3}}$

f $\frac{5}{2 - 3\sqrt{2}}$

True or False?

(1) $\sqrt{32} = 4\sqrt{2}$

(2) $\sqrt{40} = 10\sqrt{4}$

(3) $\sqrt{8} = \sqrt{2} + \sqrt{6}$

(4) $\sqrt{162} = 2\sqrt{9}$

(5) $\sqrt{5} \times \sqrt{5} = 5$

Adding & Subtracting Surds:

- Think of each surd as 'x' and **collect like terms** like you would in algebra! E.g.

$$4\sqrt{2} - \sqrt{2} = 3\sqrt{2}$$

Think of these
as
 $4x - 2x =$

Simplify:

$$8\sqrt{5} - 2\sqrt{3} + 4\sqrt{5} - \sqrt{3}$$

$$3\sqrt{2} + 4\sqrt{11} + 6 - \sqrt{2} - \sqrt{11} - 3$$

Harder Questions: Simplify Fully (make the exponent under the surd as small as possible)

$$\sqrt{9x^4} - \sqrt{36x^3} + \sqrt{x^3} =$$

$$10\sqrt[3]{81p^6} - \sqrt[3]{24p^6} =$$

Harder Questions: Simplify Fully (make the exponent under the sud as small as possible)

$$\begin{aligned} & \sqrt{9x^4} - \sqrt{36x^3} + \sqrt{x^3} \\ &= 3x^2 - 6\sqrt{x^2 \cdot x} + \sqrt{x^2 \cdot x} \\ &= 3x^2 - 6x\sqrt{x} + x\sqrt{x} \\ &= \boxed{3x^2 - 5x\sqrt{x}} \end{aligned}$$

$$\begin{aligned} & 10\sqrt[3]{81p^6} - \sqrt[3]{24p^6} \\ &= 10 \cdot \sqrt[3]{81} \cdot (p^6)^{\frac{1}{3}} - \sqrt[3]{24} \cdot (p^6)^{\frac{1}{3}} \\ &= 10 \cdot \sqrt[3]{27} \cdot \sqrt[3]{3} \cdot p^2 - \sqrt[3]{8} \cdot \sqrt[3]{3} \cdot p^2 \\ &= 10 \cdot 3 \cdot p^2 \cdot \sqrt[3]{3} - 2p^2 \sqrt[3]{3} \\ &= \boxed{28p^2 \sqrt[3]{3}} \end{aligned}$$

Multiplying Surds involving distributive rule

The rules for expanding brackets involving surds are identical to those for ordinary algebra.

Reminder: $a(b + c) = ab + ac$ This is known as the **distributive law**.

Example 8

Self Tutor

Simplify: **a** $\sqrt{2}(\sqrt{2} + \sqrt{3})$

b $\sqrt{3}(6 - 2\sqrt{3})$

$$\begin{aligned}\mathbf{a} \quad & \sqrt{2}(\sqrt{2} + \sqrt{3}) \\ &= \sqrt{2} \times \sqrt{2} + \sqrt{2} \times \sqrt{3} \\ &= 2 + \sqrt{6}\end{aligned}$$

$$\begin{aligned}\mathbf{b} \quad & \sqrt{3}(6 - 2\sqrt{3}) \\ &= (\sqrt{3})(6 + -2\sqrt{3}) \\ &= (\sqrt{3})(6) + (\sqrt{3})(-2\sqrt{3}) \\ &= 6\sqrt{3} + -6 \\ &= 6\sqrt{3} - 6\end{aligned}$$

Expand and simplify:

a $\sqrt{2}(\sqrt{5} + \sqrt{2})$

b $\sqrt{2}(3 - \sqrt{2})$

Expand and simplify:

a $(1 + \sqrt{2})(2 + \sqrt{2})$

b $(2 + \sqrt{3})(2 + \sqrt{3})$

Expand and simplify:

a $(1 + \sqrt{2})^2$

b $(2 - \sqrt{3})^2$

Expand and simplify:

a $\sqrt{2}(\sqrt{5} + \sqrt{2})$

$$= \sqrt{10} + \sqrt{2}\sqrt{2}$$

$$= \boxed{\sqrt{10} + 2}$$

b $\sqrt{2}(3 - \sqrt{2})$

$$= 3\sqrt{2} - \sqrt{2}\sqrt{2}$$

$$= \boxed{3\sqrt{2} - 2}$$

Expand and simplify:

a $(1 + \sqrt{2})(2 + \sqrt{2})$

$$= 2 + \sqrt{2} + 2\sqrt{2} + 2$$

$$= \boxed{4 + 3\sqrt{2}}$$

b $(2 + \sqrt{3})(2 + \sqrt{3})$

$$= 4 + 2\sqrt{3} + 2\sqrt{3} + 3$$

$$= \boxed{7 + 4\sqrt{3}}$$

Expand and simplify:

a $(1 + \sqrt{2})^2$

$$= (1 + \sqrt{2})(1 + \sqrt{2})$$

$$= 1 + \sqrt{2} + \sqrt{2} + 2$$

$$= \boxed{3 + 2\sqrt{2}}$$

b $(2 - \sqrt{3})^2$

$$= (2 - \sqrt{3})(2 - \sqrt{3})$$

$$= 4 - 2\sqrt{3} - 2\sqrt{3} + 3$$

$$= \boxed{7 - 4\sqrt{3}}$$